Executive Summary: Lithium-Ion Battery Recycling Strategic Recommendation

The Bottom Line

A cost-optimized hybrid approach combining Chinese mechanical equipment with Western safety systems achieves **40-60% CAPEX reduction** while maintaining safety and reliability standards.

Metric	Target
CAPEX Investment	180,000– 350,000
Throughput	500-1,000 kg/hour
Black Mass Yield	15-20% of input
ROI Timeline	18-36 months

Recommended Hybrid Equipment Stack

Stage	Equipment	Source	Cost	Rationale
Primary Shredder	Double-shaft shear, nitrogen-ready	China (GEP ECOTECH/ SUNY)	$15,000-35,000	40-60% cheaper than Western
Hammer Mill	High-speed impact, 1500 rpm	China (Henan Doing)	$8,000-20,000	MDPI confirms optimal RPM
Screening	Vibrating screen, multi-deck	China	$5,000-12,000	Commodity equipment
Magnetic Separator	Drum magnet, rare earth	Western (Eriez)	$8,000-20,000	Industry standard
Eddy Current Separator	Non-ferrous metal separator	Western (Eriez/ Steinert)	$15,000-35,000	Higher recovery justifies premium
Air Classifier	Density separation	China	$10,000-25,000	Chinese adequate

Stage	Equipment	Source	Cost	Rationale
Nitrogen System	Membrane generator + O2 monitoring	**Western (Generon)**	$25,000-50,000	**SAFETY CRITICAL**
Fire Suppression	Water mist + CO2 backup	**Western (Siemens/Tyco)**	$20,000-40,000	**SAFETY CRITICAL**
Gas Detection	HF, HCN, CO sensors, ATEX	**Western (Drager/ Honeywell)**	$15,000-30,000	**SAFETY CRITICAL**
Dust Collection	HEPA filtration	China + Western filters	$8,000-18,000	Commodity with premium filters

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Key Research Findings

Institution	Key Finding
Stanford	Recycling produces <50% GHG vs. mining; optimal throughput 500-2,000 kg/hr
MIT	Future: Self-assembling molecules for easier recycling
Harvard	Laser-based separation promising but lab-stage
Argonne	Mechanical processing costs 0.50−1.50/kgvs. 2-4/kg for pyrometallurgical
MDPI	1500 rpm hammer mill yields optimal 18% black mass

Cost Comparison: Three Strategies

Category	All-Western	**Hybrid (Recommended)**	All-Chinese
Total300,000-CAPEX	580,000|** 180,000-350,000**	$100,000-200,000	
Safety Risk	LOW	LOW	HIGH
Insurance	Easy	Moderate	Difficult
Reliability	HIGH	HIGH	Moderate

Why Hybrid Works:
- Safety systems: NO compromise (Western required)

- Mechanical equipment: Commoditized, Chinese = 40-60% savings
- Separation: Premium justified by improved yields (12-18 month ROI)

Top Recommended Suppliers

Chinese (Mechanical Equipment)

- **GEP ECOTECH** - Best for primary shredder, CE certified, nitrogen-purged systems
- **SUNY Group** - Good for complete lines, installation support
- **Henan Doing** - Excellent for hammer mills, responsive support

Western (Safety + Separation)

- **Generon (USA)** - Nitrogen generators, industry standard
- **Eriez (USA)** - Magnetic & eddy current separators, highest efficiency
- **Drager (Germany)** - Gas detection, ATEX certified
- **Siemens (Germany)** - Fire suppression, insurance-approved

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ROI Analysis

Scenario	Throughput	Annual Black Mass	Revenue	Gross Profit
Conservative	500 kg/hr	300 tons	900K|− 700K	
Expected	**750 kg/hr**	**810 tons**	**3.6M**|** 945K**	
Optimistic	1,000 kg/hr	1,584 tons	9.5M| 5.5M	

Key Insights:
- Minimum 750 kg/hr throughput required for positive returns
- Payback: 18-36 months (expected: 24-30 months)
- Black mass price: $3,000-6,000/ton depending on metal markets

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Perfect Solution Configuration (1,000 kg/hr)

Total CAPEX: 280,000− 350,000

Component	Cost
GEP ECOTECH double-shaft shredder (N2 purge)	$35,000
Henan Doing hammer mill, 1500 rpm	$18,000

Component	Cost
Multi-deck vibrating screen	$12,000
Eriez drum magnet	$15,000
Eriez eddy current separator	$25,000
Air classification system	$20,000
Generon nitrogen generator	$40,000
Drager gas detection	$20,000
Siemens fire suppression	$25,000
Dust collection (HEPA)	$15,000
Installation & commissioning	$55,000

Critical Success Factors

1. **Feedstock Security** - Secure 2-3 year agreements with battery collectors/OEMs before investing
2. **Black Mass Offtake** - Identify 2-3 buyers (Li-Cycle, Umicore, Glencore)
3. **Regulatory Compliance** - Budget for environmental permits and hazardous waste handling
4. **Technical Expertise** - Hire experienced recycling engineers for optimization
5. **Safety Culture** - Never compromise on safety systems; one incident can destroy the business

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Risk Warning

Risk	Mitigation
Metal Price Collapse	Monitor markets, consider hedging
Feedstock Unavailability	Long-term contracts
Safety Incident	Premium safety systems, training
Technology Change	Flexible process design
Regulatory Changes	Budget for compliance upgrades